

MoneyPoly White-Box Testing Report

Total white-box test cases: 35

Passed: 35

Failed: 0

Pylint score: 10.00/10

1. Program Overview

MoneyPoly is a command-line board game that manages players, money, board movement, property purchases, rent payments, card effects, jail rules, bankruptcy, and final winner selection. The white-box task focused on the internal logic of these workflows.

2. Codebase Audited

- `main.py`: entry point that reads player names, creates the Game object, and starts the session.
- `moneypoly/game.py`: game loop, turn flow, jail handling, movement resolution, cards, trades, mortgages, auctions, bankruptcy, and winner selection.
- `moneypoly/board.py`: board construction, tile classification, purchasable checks, and property lookup.
- `moneypoly/player.py`: player money, movement, jail state, owned properties, and net worth.
- `moneypoly/property.py`: property state, mortgage logic, rent logic, and group ownership checks.
- `moneypoly/bank.py`: bank reserves, payouts, collections, and emergency loans.
- `moneypoly/dice.py`: dice rolls and doubles tracking.
- `moneypoly/cards.py`: card data and deck behavior.
- `moneypoly/ui.py`: console display and safe input helpers.

3. Control Flow Graph Requirement

The control flow graph for submission should focus on the main execution path through `main.py`, `Game.run()`, `Game.play_turn()`, `Game._move_and_resolve()`, `Game._handle_jail_turn()`, and `Game._apply_card()`. It should show the normal turn path, jail path, three-doubles path, special-tile branches, card branches, bankruptcy elimination, and winner selection.

- Draw the CFG by hand.
- Label all nodes clearly.
- Use arrows to show the direction of control flow.
- Place the final image in whitebox/diagrams/.

4. Pylint Audit

Pylint cleanup was done iteratively so code-quality changes stayed separate from logic fixes. The final audited files score 10.00/10.

- Iteration 1: Remove unused imports and replace broad exception handling
- Iteration 2: Simplify boolean checks and tidy control-flow style issues
- Iteration 3: Add missing module, function, and class docstrings
- Iteration 4: Wrap long card definitions and normalize formatting
- Iteration 5: Refactor card handling and resolve remaining pylint design warnings
- Iteration 6: Add automated Git commit helper and track the whitebox report

5. White-Box Test Design

The test suite was designed around branch coverage, state changes, and important edge cases.

- Setup validation: rejects too few players and duplicate names.
- Dice range: verifies both dice use the full 1..6 range.
- Passing Go: verifies salary is awarded correctly on board wrap-around.
- Group rent: verifies rent only doubles on full group ownership.
- Bank consistency: verifies reserve updates for loans, collections, and payouts.
- Purchase edge cases: verifies exact-balance purchase and invalid property-state rejection.
- Rent transfer: verifies rent is both deducted from the tenant and credited to the owner.
- Mortgage and unmortgage consistency: verifies property state does not change on failed payment paths.
- Trade validation: verifies seller payment, property transfer, and invalid trade rejection.
- Jail release: verifies jail fine deduction and release path behavior.
- Property and special-tile routing: verifies buy, auction, skip, tax, free parking, and go-to-jail branches.
- Card branches: verifies collect, pay, jail, jail-free, move-to, birthday, and collect-from-all behavior.
- Railroad resolution: verifies railroad tiles resolve to actual property objects.
- Bankruptcy and turn order: verifies clean elimination, property reset, and correct

next-turn behavior.

- Net worth and winner selection: verifies asset value is counted and the richest player wins.

6. Errors Found And Fixed

Each logical issue found during white-box testing was fixed in a separate Error commit.

- Error 1: Validate game setup for minimum player count and unique names
- Error 2: Fix dice to use the full six-sided range
- Error 3: Award salary when passing or landing on Go
- Error 4: Require full ownership for group rent bonus
- Error 5: Reduce bank reserves when issuing loans and ignore negative collections
- Error 6: Handle empty card decks safely in remaining-count and repr methods
- Error 7: Allow property purchase at exact balance
- Error 8: Reject purchase of owned or mortgaged properties
- Error 9: Credit property owners when rent is paid
- Error 10: Preserve mortgage state when unmortgage payment fails
- Error 11: Prevent partial mortgage completion when the bank lacks funds
- Error 12: Transfer trade cash to the seller
- Error 13: Reject invalid trade requests such as negative cash and self-trade
- Error 14: Deduct the jail fine from the player on voluntary release
- Error 15: Reject auctions for owned or mortgaged properties
- Error 16: Add railroad property objects so railroad tiles are resolvable
- Error 17: Call the interactive menu before rolling each normal turn
- Error 18: Include owned properties in player net-worth calculation
- Error 19: Select the highest net-worth player as winner
- Error 20: Fix turn order after bankruptcy so the next player is not skipped
- Error 21: Cancel doubles extra-turn logic if the player was eliminated

7. Final Whitebox Audit Result

The MoneyPoly implementation in `whitebox/code/moneypoly/` is in a strong technical state. The white-box tests pass, pylint is clean, and the fix history follows the required workflow. The remaining manual submission item is the hand-drawn CFG image.